

This Risk Profile synthesizes the multi-dimensional challenges and threat vectors identified across the Department of Defense (DoD) and the National Institute of Standards and Technology (NIST) regarding the integration of Artificial Intelligence, with a specific focus on Generative AI (GAI).

1. AI Risk Definition and Framing

In the context of military and federal frameworks, **risk** is defined as the composite measure of an event's probability and the magnitude of its consequences. AI risks are inherently **socio-technical**, meaning they emerge from the interplay between technical model performance and the social context of their deployment. Unlike traditional software, AI systems are difficult to measure because they rely on training data that can change over time, and their "inscrutability" makes it hard to predict failure modes or emergent properties.

2. Core Generative AI (GAI) Risk Categories

NIST identifies 12 primary risks that are either unique to or exacerbated by Generative AI:

- **Confabulation:** The production of confident but factually incorrect or nonsensical outputs (hallucinations).
- **CBRN Information:** Future GAI may lower barriers for malicious actors to access knowledge or design tools for Chemical, Biological, Radiological, or Nuclear weapons.
- **Data Privacy:** Models may leak or infer sensitive Personally Identifiable Information (PII) through "data memorization" from training sets.
- **Information Integrity:** GAI enables highly sophisticated disinformation campaigns and "deepfakes" that can erode public trust or impact financial markets.
- **Information Security:** Lowered barriers for offensive cyber capabilities, including automated discovery of vulnerabilities and "prompt injection" attacks.
- **Harmful Bias and Homogenization:** The risk of amplifying existing disparities or creating "algorithmic monocultures" where the same model flaws impact wide sectors of society.
- **Other Risks:** Include **Intellectual Property** infringement, **Environmental Impacts** (computational costs), **Human-AI Configuration** (over-reliance or anthropomorphization), and **Value Chain Integration** (untraceable third-party components).

3. Military-Specific Operational Risks

The deployment of Large Language Models (LLMs) in defense contexts introduces high-stakes risks beyond civilian social concerns:

- **Doctrinal Non-Compliance:** Evaluation via the **ARMOR 2025** benchmark reveals that commercial LLMs frequently misinterpret foundational military doctrines such as the Law of War, Rules of Engagement (ROE), and the Joint Ethics Regulation.
 - **Operational Refusal (False Refusals):** Models often possess "over-aligned" safety filters that trigger during military necessity (e.g., refusing a lawful request involving terms like "engage" or "hostile intent"), rendering them unreliable for battlefield coordination.
 - **Decision-Making Gaps:** Models are more reliable at "Observe" tasks than "Orient" or "Decide" tasks, struggling with the normative reasoning required to justify kinetic actions.
 - **Life-Critical System Prohibition:** Current guidance strictly prohibits connecting GAI to kinetic or life-critical systems due to these reliability gaps.
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4. Socio-Technical and Strategic Risk Factors (STOPES/DISARM)

The DoD utilizes a **STOPES analysis** to categorize risks across six factors: **Social, Technological, Operational, Political, Economic, and Sustainability**. These are prioritized using the **DISARM hierarchy**, which identifies layers of dependencies:

1. **Data (Foundation):** Risks related to inaccurate collection, lack of provenance, and inherent biases.
 2. **Infrastructure:** Vulnerabilities in the sensors or networks that support AI interaction.
 3. **Security:** Factors impacting the confidentiality, integrity, and availability of AI capabilities.
 4. **Accountability:** Prerequisite for trust, ensuring systems are valid, reliable, and explainable.
 5. **Resources and Model Operations:** Correct functioning during the operational window and availability of personnel to respond to hazards.
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5. Cybersecurity Threat Vectors

DoD identifies specific attack vectors unique to the AI lifecycle:

- **Model Poisoning:** Adversaries may compromise the training environment to embed hidden backdoors or corrupt the model's logic.
 - **Data Spillage:** Inadvertent spills occurring when users input non-public DoD information or CUI into commercial, unsecured GAI services.
 - **Supply Chain Exploitation:** Compromised hardware or third-party datasets used in model development can introduce vulnerabilities early in the design phase.
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6. Bureaucratic and Structural Friction Points

Task Force Lima and other analysts identify institutional barriers that create risk through **inaction or slow adoption**:

- **Procurement Incompatibility:** Traditional, hardware-centric acquisition processes hinder the continuous development and TEVV (test, evaluation, verification, and validation) cycles required for AI.
- **Data Silos and Rights:** Complexity in negotiating data rights between DoD components and industry partners creates operational bottlenecks.
- **Talent and Infrastructure:** A shortage of technical talent and limited access to high-security (secret/top-secret) cloud compute resources prevent the scaling of secure AI tools.

To manage this profile, the DoD emphasizes **Responsible AI (RAI)** practices through the five ethical principles: **Responsible, Equitable, Traceable, Reliable, and Governable**.

Would you like me to create a **tailored report** based on this risk profile that includes specific mitigation strategies and checklists for AI program managers?

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